ASSIGNMENT 4 - CO4

TASK 6 - STORAGE DESIGN FOR A VIDEO-STREAMING START-UP

13 Marks | Bloom's Level: K5 - Evaluate

1. Stage 1 - Interpret the Workload

The start-up streams lecture videos to 200,000 students. Each video is 300 MB to 2 GB, is uploaded once and is read many thousands of times, mostly during the first week. Students may jump to any minute. This is a write-once/read-many workload with high sequential delivery, random seek/start operations, a first-week popularity skew and geographically distributed viewers.

The question does not state catalogue size, bitrate, concurrent-viewer percentage, monthly views, target latency, segment duration or cloud region. This report therefore uses an explicit planning scenario. These assumptions are not requirements given by the paper and must be replaced with production analytics.

Requirement	Chosen measurable target (assumption)
Peak concurrency	4,000 viewers (2% of 200,000 students).
Average bitrate	8 Mbit/s per viewer with adaptive bitrate for weaker links.
Peak aggregate throughput	32 Gbit/s payload; 38.4 Gbit/s with 20% headroom.
Playback startup	p95 ≤ 2 s campus; p95 ≤ 4 s remote.
Seek-to-play	p95 ≤ 2 s after jumping to any minute.
Playback stability	Rebuffering ≤ 1% of total playback time at the defined peak.
Availability	99.9% monthly target; tolerate one storage-node failure.

Collect request arrival time, video/segment ID, cache hit or miss, time to first byte/frame, rebuffer duration, selected bitrate and client region. These measurements validate the assumptions and expose hot-set changes during the first week.

2. Stage 2 - Propose the Storage Architecture

Use a three-node Linux origin/storage cluster. HDD RAID 10 stores the durable video catalogue; mirrored NVMe SSDs cache the hottest videos and recently requested segments. Store two copies of every video on separate nodes. A request router selects a healthy, less-loaded node, while a manifest/time-index service maps a requested minute to its media segment.

Delivery path: Student player → request router/manifest service → SSD cache hit or HDD origin read → segmented media response. The three nodes distribute replicas and cache hot segments; a separate monitoring process tracks load, latency and cache-hit ratio.

Component	Proposed configuration
Origin nodes	3 Linux nodes; each has 8 × 8 TB HDDs in RAID 10, giving 32 TB usable per node.
SSD cache	Each node has 2 × 1.92 TB NVMe in RAID 1, giving 1.92 TB usable cache per node.
Filesystem	XFS on Linux origin nodes; extent-based allocation suits large media files. ext4 is an acceptable alternative if it is the supported operational standard.
Allocation	Extent-based allocation for sequential video segments; manifest/time index for random minute seeking.
Replication	Two copies across separate nodes/failure domains. Local RAID 10 provides additional disk-level redundancy.
Network	25 GbE per node; 75 Gb/s aggregate and 50 Gb/s after one node/link failure.

Package the videos into HTTP streaming segments and manifests (for example, HLS-style playlists). A playlist identifies media segments and durations; the client downloads the chosen segment and continues in sequence. Segmenting makes seeking and adaptive bitrate selection practical while the uploaded master video remains immutable.

3. Stage 3 - Critical Design Decisions

Decision	Selection	Reason
Filesystem block size	4 KiB	Common Linux block size; supports metadata and extent mapping without excessive internal fragmentation for small files.
Streaming I/O	1 MiB read-ahead; 4-second segments	Larger sequential reads reduce request overhead; a seek begins at the segment selected by the manifest.
HDD scheduler	mq-deadline where available and validated	Can manage read latency in HDD queues; confirm availability in the actual kernel.
NVMe scheduler	none where supported and validated	Avoids unnecessary host-side reordering for fast parallel SSD queues; verify support on the actual device/kernel.
Cache policy	Keep top 10% videos and recently popular first-week segments; LFU/LRU hybrid.	Planning assumption: this hot set is expected to satisfy 85% of requests. Monitor and tune the actual hit ratio.
Replication factor	2 copies across separate nodes	Tolerates one node loss while avoiding the cost of three full copies; RAID 10 adds local disk fault tolerance.

The 85% cache hit ratio is a simulation input, not a measured fact. The cache controller should track first-week view counts and promote videos/segments with sustained demand. Prefetch the next segments for active sessions and avoid evicting segments still heavily used.

4. Stage 4 - Design and Calculate

4.1 Catalogue capacity

For a planning example of 1,000 videos averaging 1 GiB (within the stated 300 MB–2 GB range): **one copy = 1,000 GiB = 0.977 TiB; two replicas = 2,000 GiB = 1.953 TiB.** For 1,000 videos at the lower size limit, capacity is 300 GB; at the upper limit it is 2,000 GiB (1.953 TiB). The actual catalogue size scales linearly with video count.

Each node has eight 8 TB HDDs in RAID 10: usable capacity = (8/2) × 8 TB = 32 TB. Three nodes provide 96 TB of post-RAID capacity; with two replicas, theoretical logical capacity is 48 TB before metadata, reserved space and rebuild headroom. Each node's two 1.92 TB NVMe drives in RAID 1 provide 1.92 TB usable cache; all three nodes provide 5.76 TB cache.

4.2 Peak throughput

Peak viewers = 200,000 × 2% = 4,000. With an assumed 8 Mbit/s stream: **4,000 × 8 Mbit/s = 32 Gbit/s = 4 GB/s.** Add 20% headroom: **32 × 1.20 = 38.4 Gbit/s.** Three 25 GbE links supply 75 Gbit/s aggregate; following one node/link failure, two links supply 50 Gbit/s, above the 38.4 Gbit/s target. CPU, switches, storage, replication and failover must still be benchmarked.

Traffic path	Share	Throughput at 32 Gbit/s
NVMe hot cache (85% hit assumption)	85%	27.2 Gbit/s (3.4 GB/s)
HDD origin (15% misses)	15%	4.8 Gbit/s (0.6 GB/s)
Total	100%	32 Gbit/s (4 GB/s)

4.3 Disk reads to start playback at any minute

Use 4 KiB filesystem blocks, 4-second media segments and a 1 MiB initial read-ahead. At 8 Mbit/s, a 4-second segment is about 4 MB. There are 15 segments per minute; the time index maps minute m to approximately segment index 15m. Keep the manifest/time index in memory.

Seek state	Logical I/O requests before playback data is available
Warm metadata/index	1 media-data read request for the initial 1 MiB read-ahead. This spans 256 × 4 KiB filesystem blocks.
Cold metadata/index	1 metadata/index read + 1 media-data read = up to 2 logical I/O requests.

A 4 MB segment occupies about 977 × 4 KiB blocks. The player need not wait for the entire segment: it prefetches 1 MiB and continues fetching sequentially. HLS playlists identify media-segment URIs and durations; the client can select the segment from which playback begins.

5. Stage 5 - Implement and Demonstrate

The included C program compares two alternatives using the same deterministic request-arrival trace. It simulates parallel storage service lanes and measures per-request response time. This is a queue/service simulation, not a

physical HDD/SSD benchmark.

Simulation input	Value
Playback/segment requests	10,000
Arrival rate	About 1,000 requests/second (inter-arrival 0.5–1.5 ms).
Design A	24 HDD service lanes; random-read service time 8–10 ms.
Design B	4 SSD cache lanes at 0.25–0.40 ms for cache hits; 24 HDD lanes for misses.
Hot-cache hit ratio	85% assumed; 15% miss to HDD.

Metric	A: HDD-only	B: SSD cache + HDD
Requests completed	10,000	10,000
Mean response	9.007 ms	1.631 ms
p50 response	9.011 ms	0.339 ms
p95 response	9.904 ms	9.360 ms
p99 response	9.979 ms	9.886 ms
Maximum response	10.000 ms	10.000 ms
Cache hits / misses	Not applicable	8,496 / 1,504

The hybrid design gives a 5.52× lower average response and an 81.9% reduction in average response. Its p50 is much lower because most requests hit SSD. However, p95 improves only from 9.904 ms to 9.360 ms: the 15% HDD-miss tail remains. To improve tail latency, increase hot-set coverage beyond 95%, add SSD cache capacity, or promote popular first-week segments earlier.

These figures are actual outputs from the included C program in the preparation environment, not measurements from the user's physical drives. The simulator does not model TCP, HTTP, codecs, real cache replacement, disk firmware or full OS scheduling. Production acceptance still requires physical `fio` and end-to-end player tests.

6. Alternative Concept - Cloud Object Storage + CDN

Store the catalogue as cloud objects and deliver video segments through a CDN. The player loads the manifest, requests the segment for the selected minute and continues fetching subsequent segments. CDN cache hits serve from an edge location near the viewer and reduce repeated origin retrieval.

6.1 Playback startup by city/edge distance

Illustrative model: DNS/TLS setup = 0.25 s; startup buffer = 2 seconds at 8 Mbit/s = 2 MB; viewer link = 20 Mbit/s, so buffer transfer = 0.8 s. Startup = setup + RTT to edge + buffer transfer.

Example path	Assumed RTT	Estimated cache-hit startup
City A - near edge	20 ms	0.25 + 0.020 + 0.80 = 1.07 s
City B - regional edge	50 ms	0.25 + 0.050 + 0.80 = 1.10 s
City C - farther edge	100 ms	0.25 + 0.100 + 0.80 = 1.15 s

These are model estimates, not actual latencies for named cities. A cache miss may add an assumed 0.3–0.8 s origin-fetch delay, producing about 1.37–1.95 s under the same assumptions. Actual tests must measure ISP routing, RTT, bandwidth, DNS/TLS reuse and edge cache status. A CDN improves cacheable delivery by serving near-edge content and avoiding a full origin round trip.

6.2 Monthly storage and transfer cost

Assume each of the 200,000 students watches one average 1 GiB video per month: **200,000 × 1 GiB = 195.3 TiB, approximately 214.7 TB decimal per month.** If 80% of viewing occurs in the first week, about 156.25 TiB (171.8 TB decimal) is transferred in that week. A 1 GiB video split into approximately 4 MB segments creates about 269 requests; 200,000 views create approximately 53.8 million viewer segment requests before cache effects.

AWS's CloudFront pricing page currently lists a Premium base plan at USD 1,000/month with 50 TB transfer, 5 TB S3 storage and 500 million requests; it states that transfer may be configured up to 600 TB for a new flat-rate price. The 214.7 TB/month example exceeds the standard 50 TB allowance; obtain a configured-tier quote. The catalogue is within the listed 5 TB S3 allowance, subject to storage-class and plan terms.

USD 1,000 is the listed base price, not a final quote for this workload.

The final monthly bill must include the configured CDN transfer tier, any storage/request/management charges not covered by the plan, taxes and other services. Since the assignment gives neither monthly views nor a cloud region, this is a transparent planning estimate rather than a price quotation.

6.3 Operating effort

Cloud delivery reduces hardware purchasing, rack/power work and drive replacement. It still requires video packaging/transcoding, bucket policies and IAM, signed URLs, CDN cache rules, versioning/invalidation, monitoring, cost alerts, lifecycle and retention policies, backups and origin-failure planning. Work shifts from hardware operations toward cloud security, observability and cost management.

7. Final Recommendation

For a budget limited to a few servers, use the three-node hybrid origin: HDD RAID 10 for durable capacity, NVMe for hot first-week segments, two replicas across nodes, extent-based storage and segmented HTTP delivery with a cached time index. This supports sequential playback and minute-level seeks while leaving room to grow the SSD cache. The simulation shows a strong average-latency gain but also shows that cache misses dominate the tail.

Use cloud object storage plus a CDN when geographic reach and reduced hardware operations justify the transfer bill. A hybrid evolution is practical: retain the origin catalogue on the three nodes and add a CDN for campus/home distribution, monitored against cache-hit and egress targets.

8. References

[1] RFC 8216, *HTTP Live Streaming* - playlists, media segments and client segment selection. https://www.rfc-editor.org/rfc/rfc8216.html

[2] AWS, *Amazon CloudFront CDN pricing* - flat-rate plans, monthly price, transfer allowances, S3 storage and request limits. https://aws.amazon.com/cloudfront/pricing/ (checked 5 November 2026).

[3] AWS, *What is Amazon CloudFront?* - edge locations, caching and origin forwarding. https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/ (checked 5 November 2026).

Source alignment: Assignment 4 - CO4, Task 6 - Design Challenge, 13 Marks, pages 39–41 of the supplied question paper.